

FORMULARIO TRIGONOMÉTRICO ALGEBRA Y TRIGONOMETRÍA MÓDULO 2



Valores exactos de:						Fórmulas suma y resta de ángulos		Fórmulas reducción primer cuadrante	
	α	$\pi/6$	$\pi/4$	$\pi/3$		$\begin{aligned}\sin(\alpha \pm \beta) &= \sin \alpha \cos \beta \pm \cos \alpha \sin \beta \\ \cos(\alpha \pm \beta) &= \cos \alpha \cos \beta \mp \sin \alpha \sin \beta \\ \tan(\alpha \pm \beta) &= \frac{\tan \alpha \pm \tan \beta}{1 \mp \tan \alpha \tan \beta}\end{aligned}$		$\begin{aligned}\sin(\pi \pm \alpha) &= \mp \sin \alpha; & \sin(2\pi - \alpha) &= -\sin \alpha \\ \cos(\pi \pm \alpha) &= -\cos \alpha; & \cos(2\pi - \alpha) &= \cos \alpha \\ \tan(\pi \pm \alpha) &= \pm \tan \alpha; & \tan(2\pi - \alpha) &= -\tan \alpha\end{aligned}$	
sen	1/2	$\sqrt{2}/2$	$\sqrt{3}/2$						
cos	$\sqrt{3}/2$	$\sqrt{2}/2$	1/2						
tan	$1/\sqrt{3}$	1	$\sqrt{3}$						
Valores para ángulos cuadrantales						Fórmulas ángulos doble		Fórmulas reducción de cofunciones	
	0	$\pi/2$	π	$3\pi/2$	2π	$\begin{aligned}\sin(2\alpha) &= 2 \sin \alpha \cos \alpha \\ \cos(2\alpha) &= 2 \cos^2 \alpha - 1 = 1 - 2 \sin^2 \alpha \\ \tan(2\alpha) &= \frac{2 \tan \alpha}{1 - \tan^2 \alpha}\end{aligned}$		$\begin{aligned}\sin\left(\frac{\pi}{2} - \alpha\right) &= \cos \alpha; & \cos\left(\frac{\pi}{2} - \alpha\right) &= \sin \alpha \\ \sec\left(\frac{\pi}{2} - \alpha\right) &= \csc \alpha; & \csc\left(\frac{\pi}{2} - \alpha\right) &= \sec \alpha \\ \tan\left(\frac{\pi}{2} - \alpha\right) &= \cot \alpha; & \cot\left(\frac{\pi}{2} - \alpha\right) &= \tan \alpha\end{aligned}$	
sen	0	1	0	-1	0				
cos	1	0	-1	0	1				
tan	0	$\nexists$	0	$\nexists$	0				
csc	$\nexists$	1	$\nexists$	-1	$\nexists$				
sec	1	$\nexists$	-1	$\nexists$	1				
cot	$\nexists$	0	$\nexists$	0	$\nexists$				
Signo funciones Circulares						Fórmulas ángulos medios		Fórmulas reducción de cofunciones para $\pi/2 + \alpha$	
	I	II	III	IV		$\begin{aligned}\sin \frac{\alpha}{2} &= \pm \sqrt{\frac{1 - \cos \alpha}{2}} \\ \cos \frac{\alpha}{2} &= \pm \sqrt{\frac{1 + \cos \alpha}{2}} \\ \tan \frac{\alpha}{2} &= \pm \sqrt{\frac{1 - \cos \alpha}{1 + \cos \alpha}} = \frac{\sin \alpha}{1 + \cos \alpha}\end{aligned}$		$\begin{aligned}\sin\left(\frac{\pi}{2} + \alpha\right) &= \cos \alpha; & \cos\left(\frac{\pi}{2} + \alpha\right) &= -\sin \alpha \\ \sec\left(\frac{\pi}{2} + \alpha\right) &= -\csc \alpha; & \csc\left(\frac{\pi}{2} + \alpha\right) &= \sec \alpha \\ \tan\left(\frac{\pi}{2} + \alpha\right) &= -\cot \alpha; & \cot\left(\frac{\pi}{2} + \alpha\right) &= -\tan \alpha\end{aligned}$	
sen	+	+	-	-					
cos	+	-	-	+					
tan	+	-	+	-					
Identidades básicas						Funciones circulares pares e impares		Teorema de los Senos	
$\begin{aligned}\sin \alpha \cdot \csc \alpha &= 1 \\ \cos \alpha \cdot \sec \alpha &= 1 \\ \tan \alpha \cdot \cot \alpha &= 1\end{aligned}$						$\begin{aligned}\sin(-\alpha) &= -\sin \alpha \\ \cos(-\alpha) &= \cos \alpha \\ \tan(-\alpha) &= -\tan \alpha\end{aligned}$		$\frac{\sin \alpha}{a} = \frac{\sin \beta}{b} = \frac{\sin \gamma}{c}$	
Identidades fundamentales						Período de funciones circulares		Teorema de los Cosenos	
$\begin{aligned}\sin^2 \alpha + \cos^2 \alpha &= 1 \\ \tan^2 \alpha + 1 &= \sec^2 \alpha \\ \cot^2 \alpha + 1 &= \csc^2 \alpha\end{aligned}$						$\begin{aligned}\sin(\theta + 2k\pi) &= \sin \theta, & k \in \mathbb{Z} \\ \cos(\theta + 2k\pi) &= \cos \theta, & k \in \mathbb{Z} \\ \tan(\theta + k\pi) &= \tan \theta, & k \in \mathbb{Z}\end{aligned}$		$\begin{aligned}a^2 &= b^2 + c^2 - 2bc \cos \alpha \\ b^2 &= a^2 + c^2 - 2ac \cos \beta \\ c^2 &= a^2 + b^2 - 2ab \cos \gamma\end{aligned}$	